Databases for Big Data

Exercises: Query Processing

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Example database Sales

The following database contains two entity tables, Customer and Product, which store descriptive information about business entities. A Sales relation links these entities and records measurable events (e.g., quantities sold), forming a simple OLAP schema suitable for analytical queries.

```
CREATE TABLE Customer (
    cid      INT PRIMARY KEY,
    name     TEXT NOT NULL,
    region   TEXT
);

CREATE TABLE Product (
    pid        INT PRIMARY KEY,
    name       TEXT NOT NULL,
    category   TEXT
);

CREATE TABLE Sales (
    sale_id  INT PRIMARY KEY,
    cid      INT NOT NULL,
    pid      INT NOT NULL,
    quantity INT NOT NULL,
    FOREIGN KEY (cid) REFERENCES Customer(cid),
    FOREIGN KEY (pid)  REFERENCES Product(pid)
);
```

Here are the examples of the tables that are instances of the above relations.

Product

pid	name	category
1	Laptop	Electronics
2	Desk Chair	Furniture
3	Coffee Maker	Appliances
4	Notebook	Stationery
5	Headphones	Electronics
6	Desk Lamp	Furniture
7	Water Bottle	Accessories

Customer

cid	name	region
1	Alice Brown	West
2	Bob Smith	East
3	Carla Torres	North
4	Daniel Kim	South
5	Eva Johnson	Central

Sales

sale_id	cid	pid	quantity
1	1	1	2
2	1	4	5
3	2	3	1
4	2	7	3
5	3	2	2
6	3	5	1
7	4	1	1
8	4	6	2
9	5	3	4
10	5	7	2

Queries

Query 1.

```
select P.pid, P.category
from Product P, Sales S
where P.name = 'Notebook'
and   P.pid=S.pid
and   S.quantity>10;
```

Query 2.

```
select C.name, P.name, S.quantity
from Product P, Sales S, Customer C
where C.cid=S.cid
and   P.pid=S.pid
and   C.region='West';
```

Exercises: Queries

Exercise 1.

Present the query tree for Query 1 and Query 2.

Exercise 2.

Present the pipelines of Query 1 and Query 2.

Exercise 3.

The SQL query

select sale_id
from Sales
where pid=1 or cid=22;

scans and filters the relation Sales. Write a pseudocode for a branchless scan.

Exercise 4.

Sketch the execution of Query 1 and Query 2 when using the following processing models.

a) Iterator model
b) Materialization model
c) Vectorized / Batch model

Descriptive and explanatory questions.

1. Present the ways in which MonetDB optimizes the execution of queries.

2. Describe the concepts of a query plan, an operator, an operator instance, a task and a task set in MonetDB.

3. Describe the method for converting the code with branches into the branchless code.

4. Present the iterator query processing model.

5. Present the materialization query processing model.

6. Present the vectorized / batch query processing model.

7. Describe the mechanisms of the push-based iterator model.

8. Present the classification of parallelisms that are used in DBMS-s. Give an example of each type of parallelism.